

Daily Quiz - 6 (ELL782/ELL7282) - Computer Architectures

Instructions

- **Time:** 5 minutes
- Maximum marks: 10
- Number of questions: 10
- Each question carries **1 mark**.
- Questions marked **MCQ** have exactly one correct option.
- Questions marked **MSQ** may have one or more correct options. For an MSQ, select all correct options and no incorrect option.
- There is **no partial credit** for an MSQ.

Points: 10/10

Name * 

test

Entry Number * 

test

✓ **Correct** 1/1 Points

A call instruction is located at address 0x0400, and the called function begins at address 0x0520. What values are written to ra and pc by the call instruction? 

☐ ra = 0x0404, pc = 0x0524

☐ ra = 0x0400, pc = 0x0520

☒ ra = 0x0404, pc = 0x0520

☐ ra = 0x0520, pc = 0x0404

✓ **Correct** 1/1 Points

Which statements about function calls and activation blocks are correct? 

Select all that apply.

- ☒ An activation block may contain arguments, a return address, a register-spill area, and local variables.
- ☒ In a caller-saved scheme, the caller saves the registers whose values it needs after the function call.
- ☒ A stack is suitable for storing activation blocks because function calls and returns naturally follow a last-in, first-out order.
- ☐ In a callee-saved scheme, the caller must restore every register modified by the callee.

✓ **Correct** 1/1 Points

The recursive factorial routine allocates an 8-byte activation block for every non-base invocation. If factorial(4) starts with: sp = 0x1000 what is the smallest value reached by sp before the recursion starts returning? 

☐ 0x0FF0

☒ 0x0FE8

☐ 0x0FE0

☐ 0x0FF8

✓ **Correct** 1/1 Points

In the recursive factorial routine, why are the following save and restore operations necessary?

st ra, 4[sp] and,
after the recursive call: ld ra, 4[sp] 

- ☐ The cmp instruction stores its comparison result in ra.
- ☐ The stack pointer cannot be modified unless ra is first written to memory.
- ☐ The ret instruction always clears ra before jumping.
- ☒ A recursive call overwrites ra; without saving it, the current invocation can lose its own return address.

✓ **Correct** 1/1 Points

A downward-growing stack initially has: $sp = 0x8000$ A caller allocates 12 bytes for its activation block. Its callee then allocates 20 bytes. What are the values of sp:

1. while the callee is active, and
2. immediately after only the callee deallocates its activation block?



- ☐ (0x7FF4, 0x7FE0)
- ☒ (0x7FE0, 0x7FF4)
- ☐ (0x7FEC, 0x8000)
- ☐ (0x7FE0, 0x8000)

✓ **Correct** 1/1 Points

A call instruction is located at: $PC = 0x1000$ and its target address is $0x1080$. For a branch-format instruction, the target address is computed as: Target address = $PC + offset \times 4$ Which pair (ra, offset) is correct? 

- ☐ (0x1004, 128)

- ☐ (0x1000, 128)
- ☐ (0x1080, 32)
- ☒ (0x1004, 32)

✓ **Correct** 1/1 Points

For the register-format instruction: `sub r9, r6, r3` which sequence correctly gives the fields in the following order? opcode | I | rd | rs1 | rs2



- ☐ 00001 | 1 | 1001 | 0110 | 0011
- ☐ 00001 | 0 | 0110 | 1001 | 0011
- ☒ 00001 | 0 | 1001 | 0110 | 0011
- ☐ 00000 | 0 | 1001 | 0011 | 0110

✓ **Correct** 1/1 Points

Which statements about the 32-bit SimpleRisc instruction encoding are correct?



Select all that apply.

- ☒ Each register field is 4 bits wide because SimpleRisc has 16 registers.
- ☒ The branch format uses a 27-bit PC-relative offset, and the effective displacement is $\text{offset} \times 4$ bytes.
- ☐ I = 1 means that the second operand is taken from a register.
- ☒ The opcode field is 5 bits wide.

✓ **Correct** 1/1 Points

Which field assignment is correct for the instruction: `ld r5, 20[r3]`

- ☒ opcode = 01110, I = 1, rd = 0101, rs1 = 0011, immediate = 20
- ☐ opcode = 01110, I = 1, rd = 0011, rs1 = 0101, immediate = 20
- ☐ opcode = 01111, I = 1, rd = 0011, rs1 = 0101, immediate = 20
- ☐ opcode = 01110, I = 0, rd = 0101, rs1 = 0011, rs2 = 20

✓ **Correct** 1/1 Points

The fields of a register-format instruction are:

opcode = 00010

I = 0

rd = 0100

rs1 = 0011

rs2 = 0101

Which assembly instruction do these fields represent? 

- ☐ mul r3, r4, r5
- ☒ mul r4, r3, r5
- ☐ sub r4, r3, r5
- ☐ mul r4, r5, r3

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